



Multispectral Sensors in Remote Sensing

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Multispectral Sensors

Introduction:

- Multispectral sensors capture image data at specific wavelength bands across the electromagnetic spectrum.
- Widely used in Earth Observation for detailed analysis of land, water, and atmospheric phenomena.
- They differ from panchromatic (single band) and hyperspectral (hundreds of narrow bands) sensors.

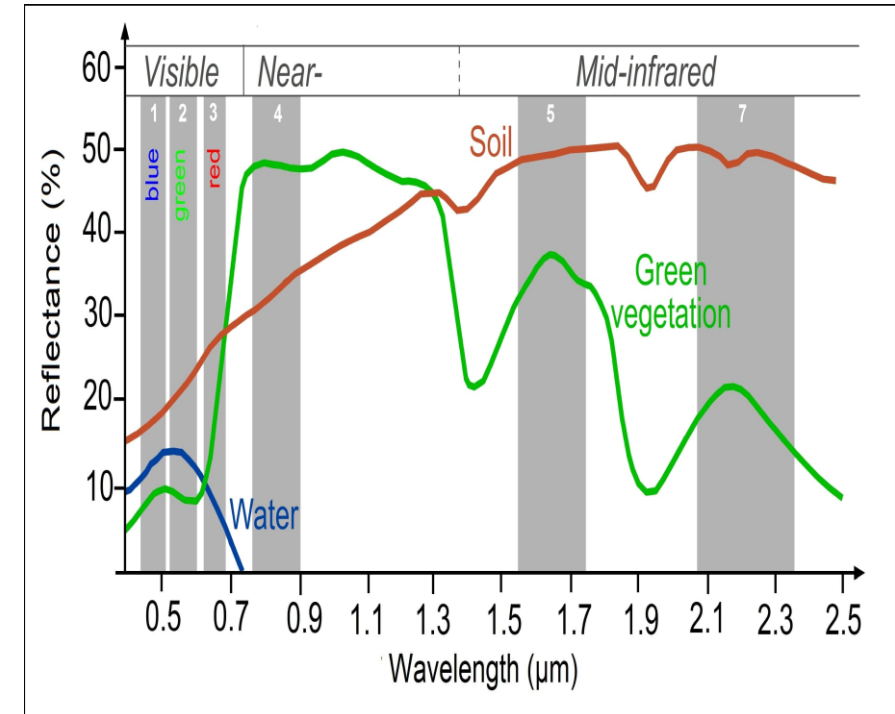
Key Concepts:

- **Electromagnetic spectrum regions:** Visible (VIS), Near-Infrared (NIR), Shortwave Infrared (SWIR), Thermal Infrared (TIR).
- **Sensor Types:** Passive optical (collecting naturally reflected sunlight), active sensors (emit their own signals).



Multispectral Sensors

- **Scanning Mechanisms:**
 - **Whiskbroom (across-track):** Uses moving mirrors; precise, but slower coverage.
 - **Pushbroom (along-track):** Linear array detects data across a swath in one go.
- **Important Parameters:**
 - **Spatial resolution** (detail of ground features),
 - **Spectral resolution** (number/width of bands),
 - **Temporal resolution** (revisit frequency).



Spectral Reflectance Curve



MSS: Multispectral Scanner

- **Satellites:** Landsat 1, 2, 3, 4, 5
- **Agency:** NASA/USGS (USA)
- **Launched:** 1972-2013
- **Type:** Passive OIR, Scanning-Imaging (Whiskbroom)
- **Spatial Resolution:** 79 m (VIS/NIR), 240 m (TIR)
- **Spectral Resolution:** 4 bands, 5 bands (for Landsat 3)
- **Temporal:** 18 days
- **Radiometric:** 6 bits
- **Swath:** 185 km
- **Impact:** Revolutionized ability to monitor large areas.

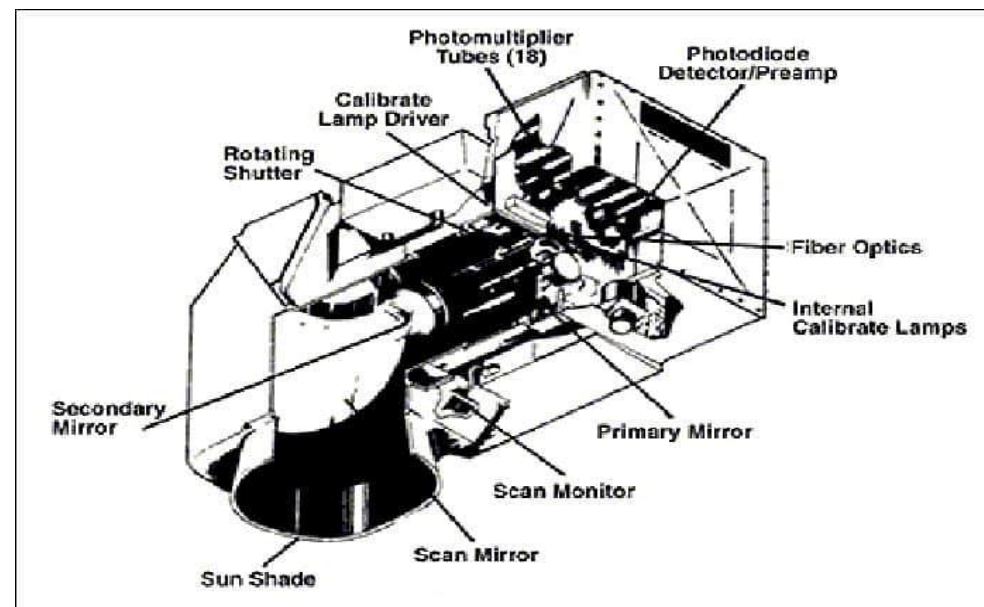


Figure: Cutaway view of the MSS (image credit: NASA)



MSS: Multispectral Scanner

Band Name	Wavelength	Applications
Green	0.5-0.6 μm	mapping coastal features in sediment-laden water
Red	0.6-0.7 μm	mapping roads and urban areas
NIR-1	0.7-0.8 μm	vegetation studies
NIR-2	0.8-1.1 μm	vegetation studies and mapping land/water boundaries



TM: Thematic Mapper

- **Satellites:** Landsat 4, 5
- **Agency:** NASA/USGS (USA)
- **Launched:** 1982-2013
- **Type:** Passive OIR, Scanning-Imaging (Whiskbroom)
- **Spatial Resolution:** 30 m (VIS/NIR/SWIR), 120 m (TIR)
- **Spectral Resolution:** 7 bands
- **Temporal:** 16 days
- **Radiometric:** 8 bits
- **Swath:** 185 km
- **Advance:** Improved spatial, radiometric and spectral resolution over MSS..

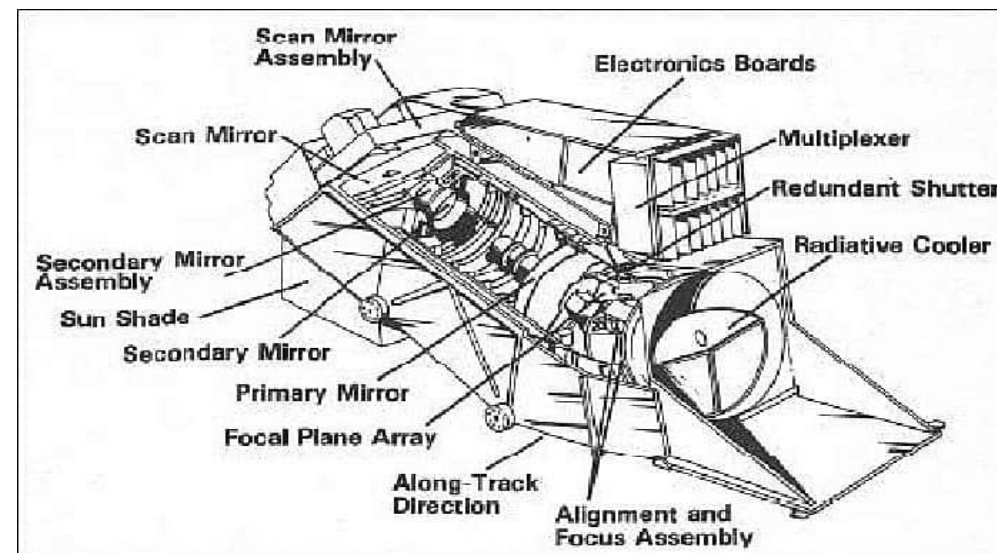


Figure: Configuration of the Thematic Mapper (image credit: NASA)



TM: Thematic Mapper

Band Name	Wavelength	Applications
Blue	0.45-0.52 μm	Coastal water mapping, soil/vegetation differentiation, deciduous/coniferous differentiation, chlorophyll absorption
Green	0.52-0.60 μm	Green reflectance, peak of healthy vegetation, plant vigor
Red	0.63-0.69 μm	Chlorophyll absorption, plant type discrimination
NIR	0.76-0.90 μm	Biomass surveys, water body delineation
SWIR-1	1.55-1.75 μm	Vegetation moisture measurement, snow/cloud differentiation
TIR	10.4-12.5 μm	Plant heat stress, thermal mapping, soil mapping
SWIR-2	2.08-2.35 μm	Hydrothermal mapping, geology



ETM+: Enhanced Thematic Mapper Plus

- **Satellites:** Landsat 7
- **Agency:** NASA/USGS (USA)
- **Launched:** 1999-2025
- **Type:** Passive OIR, Scanning-Imaging (Whiskbroom)

- **Spatial Resolution:** 30 m (VIS/NIR/SWIR), 60 m (TIR), 15 m (Pan)
- **Spectral Resolution:** 8 bands
- **Temporal:** 16 days
- **Radiometric:** 8 bits
- **Swath:** 185 km
- **Significance:** Addition of panchromatic and coastal bands, further detail.



ETM+: Enhanced Thematic Mapper Plus

Band Name	Wavelength
Blue	0.45-0.52 μm
Green	0.52-0.60 μm
Red	0.63-0.69 μm
NIR	0.78-0.90 μm
SWIR-1	1.55-1.75 μm
TIR	10.4-12.5 μm
SWIR-2	2.09-2.35 μm
Pan	0.52-0.90 μm



OLI: Operational Land Imager

- **Satellites:** Landsat 8, 9
- **Agency:** NASA/USGS (USA)
- **Launched:** 2013-Present
- **Type:** Passive OIR, Non Scanning-Imaging (Pushbroom)
- **Spatial Resolution:** 30 m (MS), 15 m (Pan)
- **Spectral Resolution:** 9 bands
- **Temporal:** 16 days
- **Radiometric:** 12 bits (Landsat 8), 14 bits (Landsat 9)
- **Swath:** 185 km
- **Significance:** Land use monitoring, water resources and coastal zone investigation, climate studies, disaster management, geological mapping.



OLI: Operational Land Imager

Band Name	Wavelength
Coastal	0.43-0.45 μm
Blue	0.45-0.51 μm
Green	0.53-0.59 μm
Red	0.64-0.67 μm
NIR	0.85-0.88 μm
SWIR-1	1.57-1.65 μm
SWIR-2	2.11-2.29 μm
Pan	0.50-0.68 μm
Cirrus	1.36-1.38 μm



TIRS: Thermal Infrared Sensor

- **Satellites:** Landsat 8, 9
- **Agency:** NASA/USGS (USA)
- **Launched:** 2013-Present
- **Type:** Passive OIR, Non Scanning-Imaging (Pushbroom)
- **Spatial Resolution:** 100 m
- **Spectral Resolution:** 2 bands
- **Temporal:** 16 days
- **Radiometric:** 12 bits (Landsat 8), 14 bits (Landsat 9)
- **Swath:** 185 km
- **Significance:** Measures land surface temperature, provides data for water irrigation management in arid regions, and tracks urban heat units.

Band Name	Wavelength
TIR-1	10.60-11.19 μm
TIR-2	11.50-12.51 μm



HRV: High Resolution Visible

- **Satellites:** SPOT 1, 2, 3
 - **Agency:** CNES (France)
 - **Launched:** 1986-2009
 - **Type:** Passive OIR, Non Scanning-Imaging (Pushbroom)

 - **Spatial Resolution:** 10 m (Pan), 20 m (MS)
 - **Spectral Resolution:** 4 bands
 - **Temporal:** 26 days
 - **Radiometric:** 8 bits
 - **Swath:** 60-117 km (Combined)
 - **Advance:** Used for land use/land cover mapping, infrastructure management, precision forestry, and military/intelligence.
- Two High-Resolution Visible (HRV) imagers are used in each satellite.
 - They operated in two modes, Panchromatic (PAN) mode and Multispectral (MS) mode, either individually or simultaneously.



HRV: High Resolution Visible

Band Name	Wavelength
Green	0.50-0.59 μm
Red	0.61-0.68 μm
NIR	0.79-0.89 μm
Pan	0.51-0.73 μm



HRVIR: High Resolution Visible and Infrared

- Platform Details

SPOT-4 Satellite (France)

Launched: March 1998

Swath Width: 60 km

Primary Uses

Vegetation discrimination and
monitoring

Water body identification

Soil moisture assessment

- Spatial Resolution

- **Spatial Resolution**

- Multispectral: **20 meters**

- Panchromatic: **10 meters**

- Ideal for regional land use studies



Spectral band configuration

■ Band	Wavelength(μm)	Application
• B1 (Green)	0.50–0.59	Vegetation health, water bodies
• B2 (Red)	0.61–0.68	Vegetation discrimination, chlorophyll absorption
• B3 (NIR)	0.79–0.89	Biomass estimation, vegetation vigour
• SWIR	1.58–1.75	Soil moisture, geological mapping



NAOMI

NAOMI: New AstroSat Optical Modular Imager

- NAOMI represents the latest advancement in high-resolution Earth observation, serving as the successor to ALOS AVNIR-2. Deployed on Japan's ALOS-3 platform in 2024, this push-broom scanner captures data in the visible and near-infrared spectrum with exceptional spatial detail.
- With its impressive 70 km swath width and uniform 3-meter resolution across all spectral bands, NAOMI excels at detailed coastal zone mapping, urban planning applications, and precision vegetation analysis.



SPECTRAL BAND CONFIGURATION

BLUE BAND

0.42–0.50 μm | 3 m resolution

Coastal water penetration

GREEN BAND

0.52–0.60 μm | 3 m resolution

Vegetation peak reflectance

RED BAND

0.61–0.69 μm | 3 m resolution

Chlorophyll absorption

NIR BAND

0.76–0.89 μm | 3 m resolution

Biomass and structure



MODIS

MODIS: Moderate Resolution Imaging Spectroradiometer

- PLATFORM

Terra Satellite (1999)

Aqua Satellite (2002)

NASA Earth Observing System

Resolution

250 m (Bands 1-2)

500 m (Bands 3-7)

1000 m (Bands 8-36)

- COVERAGE

- Swath: **2,330 km**

- Global coverage every 1-2 days

- 36 spectral bands



MODIS

APPLICATION

Vegetation Monitoring

NDVI and EVI products track global vegetation dynamics and phenology patterns

Ocean Color

Chlorophyll concentration and phytoplankton distribution in marine ecosystems

Land Surface Temperature

Day and night LST measurements for climate and urban heat island studies

Atmospheric Science

Cloud properties, aerosols, and atmospheric correction parameters



ASTER

ASTER: Advanced Spaceborne Thermal Emission and Reflection Radiometer

- ASTER (Advanced Spaceborne Thermal Emission and Reflection Radiometer) is a high spatial resolution imaging instrument aboard NASA's Terra satellite, launched in December 1999 as part of NASA's Earth Observing System (EOS).
- It provides detailed Earth imagery across 14 spectral bands spanning visible, near-infrared, shortwave infrared, and thermal infrared wavelengths, ranging from 0.52 μm to 11.65 μm . ASTER enables the creation of high-resolution maps of surface temperature, emissivity, and reflectance, and adds stereo viewing capability for Digital Elevation Model (DEM) generation.



ASTER

ASTER: Advanced Spaceborne Thermal Emission and Reflection Radiometer

SPECTRAL BAND CONFIGURATION

■ VNIR Subsystem

3 bands | 0.52–0.86 μm

Resolution: 15 meters

Surface reflectance, stereo DEM generation

■ SWIR Subsystem

■ **6 bands** | 1.6–2.43 μm

■ Resolution: 30 meters

■ Mineral identification, lithological mapping

TIR Subsystem

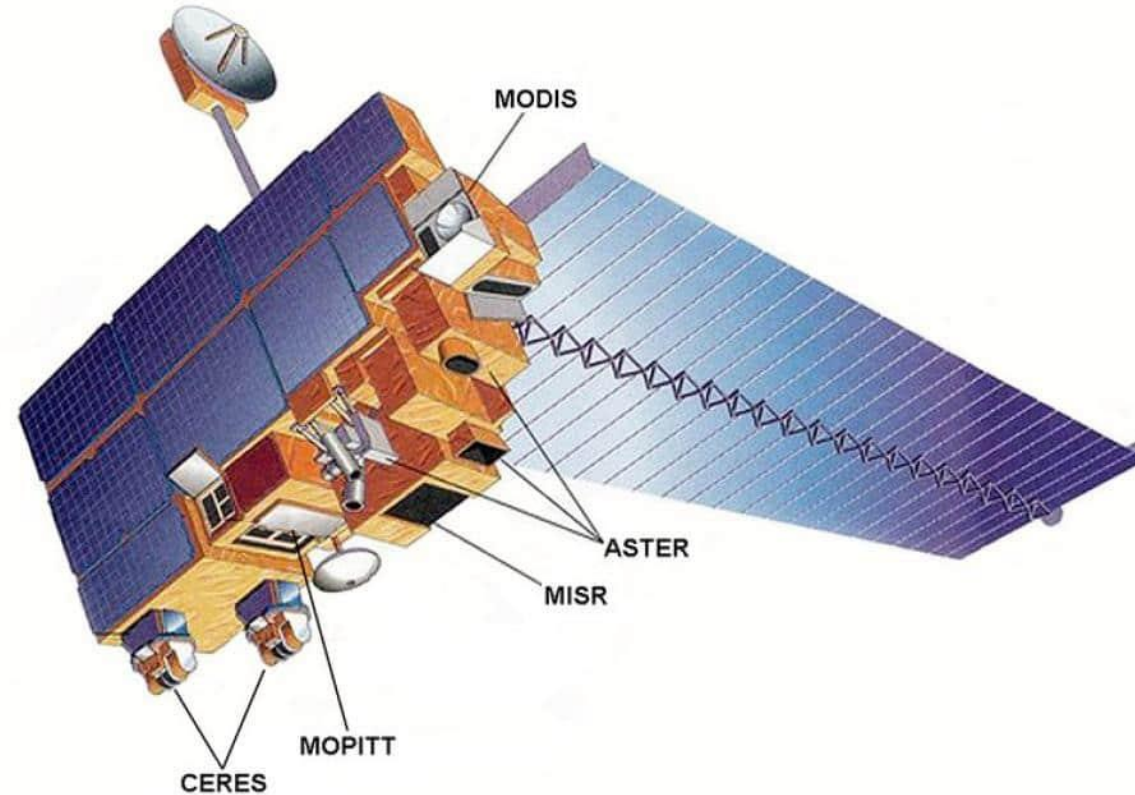
5 bands | 8.125–11.65 μm

Resolution: 90 meters

Land surface temperature, volcanic activity



Terra spacecraft showing the locations of its five instruments.





AVHRR

AVHRR: Advanced Very High Resolution Radiometer

- Operating continuously since 1978 on NOAA's polar-orbiting satellite series, AVHRR provides one of the longest records of Earth observation data. This heritage sensor has been instrumental in establishing baseline climate measurements and long-term vegetation trends with its 1.1 km spatial resolution and twice-daily global coverage.



AVHRR

AVHRR: Advanced Very High Resolution Radiometer

SPECTRAL BAND CONFIGURATION

Band	Wavelength (μm)	Primary Application
1	0.58–0.68	Cloud detection, surface mapping, vegetation discrimination
2	0.73–1.10	Vegetation indices (NDVI), biomass assessment
3A	1.58–1.64	Snow/ice discrimination, daytime cloud detection
3B	3.55–3.93	Night cloud temperature, fire detection
4	10.30–11.30	Sea surface temperature (SST), land surface temperature (LST)
5	11.50–12.50	Atmospheric water vapor correction, split-window SST



MERIS

MERIS: Medium Resolution Imaging Spectrometer

- Operating on ESA's ENVISAT platform from 2002 to 2012, MERIS was specifically designed for ocean colour remote sensing with 15 programmable spectral bands in the visible and near-infrared range (0.39–1.04 μm).
- With its exceptional 1,150 km swath width, MERIS achieved global coverage every 2-3 days, making it invaluable for monitoring dynamic ocean processes, coastal water quality, and terrestrial ecosystems.



MERIS

MERIS: Medium Resolution Imaging Spectrometer

Application

- **Ocean Colour Products:** Chlorophyll a concentration, suspended sediment, and colored dissolved organic matter (CDOM) measurements enable marine ecosystem health assessment
- **Water Quality Monitoring:** Algal bloom detection, turbidity mapping, and eutrophication assessment in coastal and inland waters
- **Atmospheric Correction:** Dedicated bands for aerosol and water vapor retrieval ensure accurate surface reflectance products
- **Coastal Zone Studies:** 300 m full resolution mode enables detailed monitoring of dynamic coastal ecosystems and land-sea interactions



MERIS

MERIS: Medium Resolution Imaging Spectrometer

*BAHAMAS, CAPTURED
BY MERIS ON 24 JAN
2004*





OSA : Optical Sensor Assembly

❑ Platform Details

IKONOS-2(USA)

Launched: September 1999

Decommissioned: March 2015

Swath Width: 11.3 km at Nadir

Primary Uses

Urban planning and management
and Land use and land cover
mapping

Monitoring of open-pit mines

❑ Spatial Resolution

- Multispectral: **3.28 meters**
- Panchromatic: **0.82 meters**
- Ideal for urban planning and management



Spectral band configuration

■ Band	Wavelength(μm)	Application
• Panchromatic	0.45 – 0.90	Used for high-resolution mapping (up to 1 meter)
• Blue	0.45 – 0.52	Useful in coastal zone studies, Water body delineation
• Green	0.52 – 0.60	Supports forestry analysis and urban green space mapping
• Red	0.63 – 0.70	Identifies vegetation species, land cover classification
• NIR	0.76 – 0.90	Vegetation health assessment, biomass estimation



QuickBird

□ Platform Details

QuickBird(USA)

Launched: October 2001

Decommissioned: January 2015

Swath Width: 16.8 km

Primary Uses

Environmental monitoring and
Natural resource exploration

Disaster management and emergency
response

□ Spatial Resolution

- Multispectral: **2.90 meters**
- Panchromatic: **0.65 meters**
- Ideal for high-resolution earth observation



SPECTRAL BAND CONFIGURATION

Band	Wavelength(μm)	Application
•Panchromatic	0.45 – 0.90	Urban infrastructure delineation and monitoring
•Blue	0.45 – 0.52	Soil and vegetation discrimination
•Green	0.52 – 0.60	Forest cover and biomass estimation
•Red	0.63 – 0.70	Land cover classification and soil discrimination
•NIR	0.76 – 0.90	Water body boundaries and sediment detection



WorldView-2

□ Platform Details

WorldView-2(USA)

Launched: October 2009

Swath Width: 16.4 km

Primary Uses

Generation of Digital Terrain
Models and Crop monitoring and
yield forecasting

Forest inventory and forest
management

□ Spatial Resolution

- Multispectral: **1.84 meters**
- Panchromatic: **0.46 meters**
- Ideal for high-resolution urban planning and infrastructure monitoring, road networks, buildings, and utilities



SPECTRAL BAND CONFIGURATION

Band	Wavelength(μm)	Application
•Panchromatic	0.45 – 0.80	High-resolution grayscale imagery for detailed mapping
•Coastal	0.40 – 0.45	Coastal and shallow water bathymetry mapping
•Blue	0.45 – 0.52	Soil and vegetation discrimination
•Green	0.52 – 0.60	Forest cover and biomass estimation
•Yellow	0.58 – 0.62	Crop maturity and stress monitoring
•Red	0.63 – 0.70	Land cover classification and soil discrimination
•Red-edge	0.70 – 0.74	Sensitive to chlorophyll content for crop health
•NIR 1	0.77 – 0.89	Vegetation biomass and health monitoring
•NIR2	0.86 – 1.04	Enhanced vegetation discrimination and biomass assessment



WorldView-3

□ Platform Details

WorldView-3(USA)

Launched: August 2014

Swath Width: 13.1 km

Primary Uses

Environmental monitoring and Mineral
and geological mapping enabled by
SWIR bands

CAVIS sensor bands that monitor clouds,
aerosols, vapors, ice, and snow

□ Spatial Resolution

- Multispectral: **1.24 meters**
- SWIR: **3.7meters**
- CAVIS: **30 meters**
- Panchromatic: **0.31 meters**
- Precise environmental and atmospheric monitoring



SPECTRAL BAND CONFIGURATION

- **Panchromatic Band (0.45 – 0.80 μm)**
High-resolution black-and-white imagery for detailed mapping and feature extraction.
Urban infrastructure and road network analysis
- **Multispectral Bands (0.40 – 1.04 μm)**
Soil and Moisture Analysis
Vegetation Monitoring
Land Cover Classification
Urban Mapping
Disaster Assessment
- **Shortwave Infrared (SWIR) Bands (1.19 - 2.36 μm)**
Mineral identification and geological mapping
Burn scar and fire damage analysis
- **CAVIS Bands (12 bands)**
Atmosphere and cloud monitoring, Aerosol and vapor detection for image correction
and snow and ice monitoring



WorldView-4

□ Platform Details

WorldView-4(USA)

Launched: January 2019

Swath Width: 13.1 km

Primary Uses

Engineering surveys especially for
construction and oil and gas facilities and

Detailed urban and infrastructure
mapping

Disaster response and recovery
assessment

□ Spatial Resolution

- Multispectral: **1.24 meters**
- Panchromatic: **0.31 meters**



SPECTRAL BAND CONFIGURATION

Band	Wavelength(μm)	Application
•Panchromatic	0.45 - 0.80	High-resolution detailed mapping and feature extraction
•Blue	0.45 - 0.51	Water body delineation and sediment analysis
•Green	0.51 - 0.58	Crop monitoring and species differentiation
•Red	0.65 - 0.69	Soil and land cover classification
•NIR	0.78 - 0.92	Water body boundaries and wetland mapping



RapidEye

□ Platform Details

RapidEye (USA)

Launched: August 2008

Decommissioned: April 2020

Swath Width: 77 km

Primary Uses

Disaster Management and Mapping and
Cartography

Land and Agriculture Monitoring

□ Spatial Resolution

- **5 meters** Orthorectified



SPECTRAL BAND CONFIGURATION

Band	Wavelength(μm)	Application
•Blue	0.44 – 0.51	Water body analysis including clarity and quality assessment
•Green	0.52 – 0.59	Urban greenery mapping and land cover classification
•Red	0.630 – 0.68	Identifying vegetation species and soil mapping
•Red-edge	0.69 – 0.73	Differentiation of vegetation species and health
•NIR	0.76 – 0.85	Forest inventory and canopy analysis



Multispectral Imager (MSI)

-MSI is an optical imaging sensor designed to capture multispectral images of the Earth. It helps in monitoring land use, vegetation, water bodies, urban areas, and disaster management. It's widely used in satellites like **Sentinel-2A and 2B**.

Sensor Type: Passive Optical

Agencies: ESA (European Space Agency) & Copernicus Programme

Orbit: Sun-synchronous, near-polar orbit

Altitude: ~786 km

Swath Width: 290 km

Spatial Resolution: 10 m, 20 m, 60 m (depending on spectral band)

Mission Timeline: Sentinel-2A (2015–Present), Sentinel-2B (2017–Present)



Multispectral Imager (MSI)

Key facts

- ❑ Captures data in 13 spectral bands ranging from visible to SWIR (Short-Wave Infrared).
- ❑ Spatial resolutions: 10m, 20m, and 60m depending on the band.
- ❑ Revisit time: Every 5 days (with both satellites in orbit).
- ❑ Features a push-broom scanner, not a whisk-broom — meaning it captures full image swaths at once.
- ❑ Great for land cover mapping, vegetation health, and coastal studies.
- ❑ Fun fact: The MSI is so precise that it can distinguish between different types of crops or tree species.



Multispectral Imager (MSI)

MSI – Multispectral Imager

Parameter	Sentinel-2A	Sentinel-2B
Launch Date	23 June 2015	7 March 2017
Orbit Type	Sun-synchronous, near-polar	Identical to Sentinel-2A (phase-shifted by 180°)
Altitude	786 km	786 km
Orbit Repeat Cycle	10 days (combined with 2B → 5 days)	10 days (phase-shifted → combined 5-day revisit)
Swath Width	290 km	290 km
Spatial Resolution	10 m / 20 m / 60 m (depending on band)	10 m / 20 m / 60 m (depending on band)
Spectral Band Configuration	13 bands (Visible, NIR, SWIR)	13 bands (Same as Sentinel-2A)



Multispectral Imager (MSI)

Sentinel-2A

Applications:

- **Agriculture:** Crop type identification, health, and yield monitoring.
- **Forestry:** Mapping forest cover, biomass, and deforestation detection.
- **Land use/land cover (LULC):** Classification and urban expansion studies.
- **Water resources:** Water quality, turbidity, and inland water mapping.
- **Disaster management:** Flood, drought, and forest fire monitoring.
- **Climate and environment:** Vegetation indices (NDVI), drought assessment, and change detection.



Multispectral Imager (MSI)

Sentinel-2B

Key Features & Differences:

Sentinel-2B ensures:

- reduced revisit time → 5 days at the equator, 2–3 days at mid-latitudes.
- Provides global multi-temporal coverage for vegetation and land monitoring.
- Data from 2A & 2B are inter-calibrated for consistency in long-term Earth observation studies.

Applications:

- ☐ crop growth assessment
- ☐ Continuous land surface monitoring
- ☐ water body mapping
- ☐ environmental disaster tracking
- ☐ climate change analysis.



Visible Infrared Imaging Radiometer Suite (VIIRS)

-The VIIRS sensor is a multispectral imaging radiometer designed to collect visible and infrared imagery of land, atmosphere, cryosphere, and oceans. It bridges the heritage of MODIS (Terra/Aqua) and AVHRR (NOAA series), providing higher spatial resolution and improved radiometric accuracy.

Sensor Type: Passive Optical (Visible, Near Infrared, Shortwave Infrared, Thermal Infrared)

Platform: Suomi NPP and NOAA-20 (JPSS Satellite Series)

Agencies: NASA (USA) & NOAA (USA)

Orbit: Sun-synchronous, near-polar orbit

Altitude: ~824 km

Swath Width: 3,040 km

Spatial Resolution: 375 m (Imagery bands), 750 m (Moderate-resolution bands)

Mission Timeline:

Suomi NPP VIIRS: Launched 28 October 2011 (Operational)

NOAA-20 VIIRS: Launched 18 November 2017 (Operational)

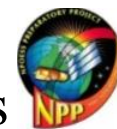
NOAA-21 VIIRS: Launched 10 November 2022 (Operational)



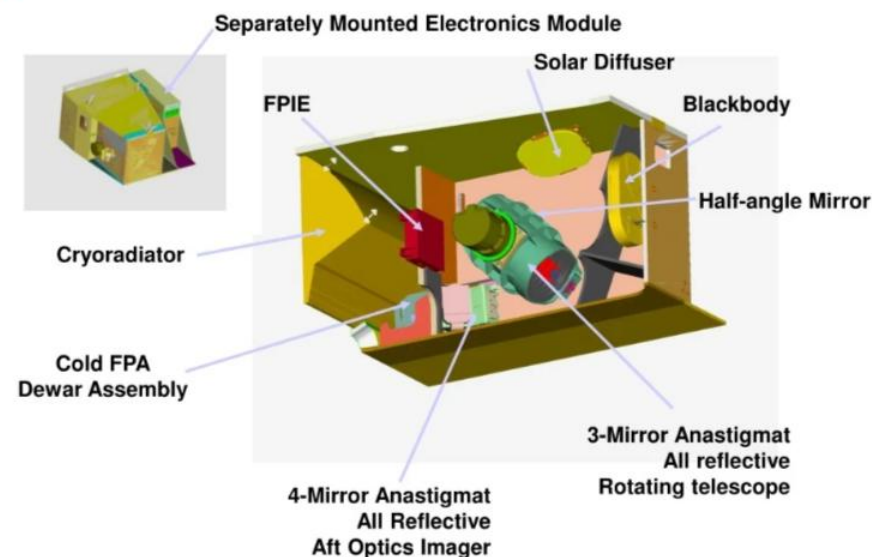
Visible Infrared Imaging Radiometer Suite (VIIRS)

Key facts

- ❑ Capable of capturing low-light nighttime imagery such as city lights and gas flares (“Black Marble” imagery).
- ❑ Has 22 spectral bands covering visible to infrared regions.
- ❑ Provides data for weather forecasting, ocean color, and fire detection.
- ❑ Detects active fires and thermal anomalies with high accuracy.
- ❑ Successor to the MODIS instrument with improved spatial resolution and calibration stability.



Visible Infrared Imaging Spectroradiometer (VIIRS)





Visible Infrared Imaging Radiometer Suite (VIIRS)

Types / Satellites Carrying VIIRS

Satellite	Launch Year	Agency	Status	Orbit Altitude	Swath Width	Revisit
Suomi NPP (S-NPP)	2011	NASA– NOAA	Active	~824 km	3,040 km	Daily
NOAA-20 (JPSS-1)	2017	NOAA	Active	~824 km	3,040 km	Daily
JPSS-2 (NOAA-21)	2022	NOAA	Active	~824 km	3,040 km	Daily



Visible Infrared Imaging Radiometer Suite (VIIRS)

Key Features:

- Combines **high radiometric sensitivity** with a wide swath for consistent daily global coverage.
- **Day/Night Band (DNB)** detects low-light emissions (e.g., city lights, aurora, wildfires, gas flares).
- Enhanced **cloud, aerosol, and ocean color detection** capability.
- Provides continuity with MODIS and AVHRR for long-term environmental data records.

Applications:

- **Land:** Vegetation indices (NDVI, EVI), urban heat mapping, land surface temperature
- **Ocean:** Chlorophyll concentration, sediment, and turbidity
- **Atmosphere:** Cloud detection, aerosol optical depth, volcanic ash tracking
- **Cryosphere:** Snow/ice monitoring, sea ice concentration
- **Disaster Monitoring:** Fire hotspots, flood mapping, dust storms, and nighttime events



Advanced Visible and Near Infrared Radiometer Type-2 (AVNIR-2)

AVNIR-2

AVNIR-2 is a **push-broom multispectral imager** designed to capture detailed land surface data in visible and near-infrared wavelengths.

It succeeded AVNIR (on ADEOS) and was mounted on **ALOS (Daichi)**.

The instrument provides high-precision optical data for land cover, coastal mapping, and disaster assessment, complementing ALOS's radar and PRISM instruments.

Sensor Type: Passive Optical (Visible to Near-Infrared)

Platform: ALOS (Advanced Land Observing Satellite)

Agency: JAXA – Japan Aerospace Exploration Agency

Orbit: Sun-synchronous, near-polar orbit

Mission Duration: 2006 – 2011

Altitude: ~692 km

Swath Width: 70 km

Spatial Resolution: 10 m (across all bands)

Viewing Angle: $\pm 30^\circ$ (pointable off-nadir capability)



Advanced Visible and Near Infrared Radiometer Type-2 (AVNIR-2)

Key facts

- ❑ Operates in 4 bands: Blue, Green, Red, and Near Infrared.
- ❑ Developed by Japan's JAXA for precise land monitoring and disaster mapping.
- ❑ Spatial resolution: 10 meters.
- ❑ Often used with ALOS's PALSAR radar data for combined optical–radar analysis.
- ❑ Useful for detecting changes in land use, vegetation, and urban expansion.
- ❑ Interesting fact: ALOS is nicknamed “Daichi,” which means “Earth” in Japanese.

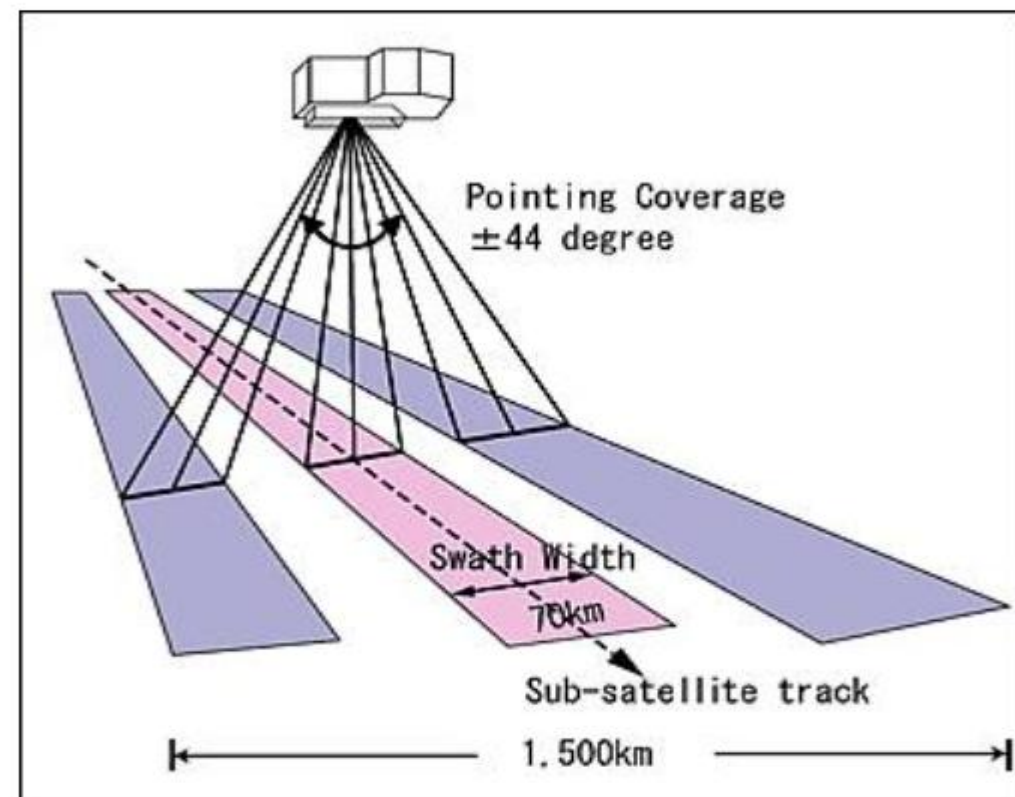


Illustration of AVNIR-2 observation capabilities. Credit: JAXA



Advanced Visible and Near Infrared Radiometer Type-2 (AVNIR-2)

Key Features

- High radiometric accuracy and geometric precision for cartographic applications.
- Forward and backward tilt capability ($\pm 44^\circ$) allowing repeated observations of the same target for stereo imaging.
- Data synergy with ALOS's PRISM (panchromatic stereo mapper) and PALSAR (L-band radar) for integrated land analysis.
- Enhanced sensitivity to vegetation, coastal waters, and urban areas.

Applications

- Land cover and land-use classification
- Vegetation monitoring and agriculture assessment
- Coastal and inland water resource studies
- Urban and regional mapping
- Disaster observation (floods, landslides, earthquakes)
- Integration with digital elevation model (DEM) generation



Linear Imaging and Self-Scanning Sensor (LISS) – IRS Series

LISS

The LISS (Linear Imaging Self-Scanning) series represents India's pioneering effort in multispectral Earth observation.

Each generation (LISS-I, LISS-II, LISS-III, LISS-IV) improved in terms of spatial, radiometric, and spectral resolution.

LISS sensors capture high-quality multispectral imagery across visible, near-infrared (NIR), and shortwave infrared (SWIR) bands.

Sensor Type: Passive Optical (Visible, Near Infrared, and Shortwave Infrared)

Platform: IRS Satellite Series (IRS-1A, IRS-1B, IRS-1C, IRS-1D, Resourcesat-1/2)

Agency: ISRO – Indian Space Research Organisation

Orbit: Sun-synchronous, near-polar orbit

Altitude: ~817 km (IRS-1A/1B)

Spatial Resolution: 5.8 m – 72.5 m (varies with version)

Swath Width: 23 km – 740 km

Mission Timeline: 1988 – Present (continuously upgraded series)



Linear Imaging and Self-Scanning Sensor (LISS) – IRS Series

Key facts

- ❑ Developed by ISRO (India), a key instrument in India's Earth observation program.
- ❑ Uses push-broom scanning technology.
- ❑ Multiple versions: LISS-I, II, III, and IV, each improving resolution and spectral range.
- ❑ Spatial resolution ranges from 23.5m (LISS-III) to 5.8m (LISS-IV).
- ❑ Commonly used for agriculture monitoring, forest mapping, and rural development.
- ❑ Interesting fact: LISS-IV provides three-band multispectral imagery with high geometric accuracy, ideal for urban mapping.



Linear Imaging and Self-Scanning Sensor (LISS) – IRS Series

Types of LISS Sensors

Series	Satellite	Resolution (m)	Swath (km)	Spectral Bands (μm)
LISS-I	IRS-1A/1B	72.5	148	0.45–0.86 (4 bands: B, G, R, NIR)
LISS-II	IRS-1A/1B	36.25	148	0.45–0.86 (4 bands)
LISS-III	IRS-1C/1D / Resourcesat	23.5	141	0.52–1.70 (G, R, NIR, SWIR)
LISS-IV	Resourcesat-1/2	5.8	23	0.52–0.86 (G, R, NIR)



Linear Imaging and Self-Scanning Sensor (LISS) – IRS Series

Features

- Push-broom scanner providing wide-area multispectral coverage.
 - Designed for resource management, crop monitoring, and environmental studies.
 - Progressive enhancement from LISS-I to LISS-IV improved radiometric accuracy and ground detail.
 - Integrated with PAN cameras for panchromatic and multispectral fusion (e.g., IRS-1C, 1D).
- Operates alongside AWiFS for large-area imaging.

Applications

- 1) Agriculture: Crop type mapping, irrigation planning, yield assessment.
- 2) Forestry: Canopy density and deforestation monitoring.
- 3) Water Resources: Reservoir and wetland mapping.
- 4) Urban Studies: Land use/land cover classification.
- 5) Disaster Management: Drought, flood, and forest fire monitoring.



CCD Camera – CBERS Series

The **Charge-Coupled Device (CCD) Camera** is the primary multispectral imaging system aboard the CBERS satellites.

It provides moderate-resolution imagery across visible to near-infrared bands, supporting environmental, agricultural, and land-use applications.

Developed jointly by Brazil and China, the CCD sensor made the **CBERS program** one of the earliest cooperative Earth observation missions.

Sensor Type: Passive Optical (Visible, Near Infrared, Panchromatic)

Platform: CBERS Satellite Series

Agencies: INPE (Brazil) & CNSA (China)

Orbit: Sun-synchronous, near-polar orbit

Altitude: ~778 km

Swath Width: 113 km

Spatial Resolution: 20 m (Multispectral), 5 m (Panchromatic)

Mission Timeline: CBERS-1 (1999–2003), CBERS-2 (2003–2007), CBERS-2B (2007–2010), CBERS-4 (2014–Present)



Key facts

- ☐ Joint collaboration between China and Brazil.
- ☐ The CCD (Charge-Coupled Device) camera captures high-resolution multispectral imagery.
- ☐ Covers 4 bands: Blue, Green, Red, and Near Infrared.
- ☐ Spatial resolution: approximately 20 meters.
- ☐ Used for environmental monitoring, deforestation studies, and land classification.
- ☐ Interesting fact: CBERS data is open-access, supporting global and regional research efforts.



CCD Camera – CBERS Series

Features

- ☐ Moderate-resolution imaging for regional environmental studies.
- ☐ Panchromatic band enables data fusion with multispectral imagery.
- ☐ Designed for land resource inventory, water management, and agriculture.
- ☐ Operational on multiple CBERS missions ensuring data continuity.
- ☐ CBERS-4 version includes improved radiometric calibration and larger dynamic range.

Applications

- ☐ Agriculture: Crop type mapping, productivity monitoring.
- ☐ Environment: Land degradation and vegetation change analysis.
- ☐ Water Resources: Lake and reservoir sedimentation studies.
- ☐ Urban Mapping: Land use/land cover classification, urban expansion.
- ☐ Disaster Management: Floodplain monitoring, deforestation tracking.



Arirang Electro-Optical Imaging Subsystem (AEISS)

The Arirang Electro-Optical Imaging Subsystem (AEISS) is a high-resolution multispectral and panchromatic imaging system developed by KARI for the KOMPSAT satellite program. It provides fine-resolution Earth observation data for national mapping, disaster monitoring, and environmental applications.

Each generation of AEISS enhanced spatial, radiometric, and spectral precision to support both civil and governmental missions.

Sensor Type: Passive Optical (Visible to Near-Infrared)

Platform: KOMPSAT (Korean Multi-Purpose Satellite) Series – KOMPSAT-2, KOMPSAT-3, KOMPSAT-3A

Agency: KARI – Korea Aerospace Research Institute

Orbit: Sun-synchronous, near-polar orbit

Altitude: ~685 km (KOMPSAT-2), ~685–690 km (KOMPSAT-3/3A)

Swath Width: 14–15 km

Spatial Resolution: 1.0 m (Panchromatic), 4.0 m (Multispectral)

Mission Timeline: KOMPSAT-2 (2006–2020), KOMPSAT-3 (2012–Present), KOMPSAT-3A (2015–Present)



Arirang Electro-Optical Imaging Subsystem (AEISS)

Key facts

- ❑ Developed by South Korea's KARI (Korea Aerospace Research Institute).
- ❑ Provides high spatial resolution: 0.5m (panchromatic) and 2m (multispectral).
- ❑ Designed for urban planning, defense, and disaster management.
- ❑ Uses optical push-broom imaging technology.
- ❑ Interesting fact: AEISS can capture color images sharp enough to identify individual vehicles from orbit.



Arirang Electro-Optical Imaging Subsystem (AEISS)

Features

- ☐ High spatial resolution suitable for urban and precision mapping.
- ☐ Capable of stereoscopic imaging for 3D terrain model generation.
- ☐ Enhanced onboard image calibration and noise reduction technology.
- ☐ Radiometric linearity ensures accurate reflectance data for quantitative analysis.
- ☐ Used for both civil mapping and defense reconnaissance operations.

Applications

- ☐ Urban Mapping: Infrastructure planning, change detection, city growth analysis.
- ☐ Agriculture: Crop health monitoring and yield estimation.
- ☐ Forestry: Biomass and canopy structure assessment.
- ☐ Disaster Management: Flood, earthquake, and landslide mapping.
- ☐ National Security: Border monitoring and reconnaissance imaging.

PlanetScope camera

Sensor Name	Vehicle	Resolution				Wavelength (μm)	Swath width (km)
		Spatial	Spectral	Temporal	Radiometric		
PlanetScope Camera (USA)	PS2 (DOVE-C) (2014-2022)	3.0-4.1 m	4 Bands	1 Day	12 Bits	0.455-0.860	25-32
	PS2.SD (DOVE-R) (2019-2022)					0.464-0.888	
	PSB.SD (SuperDOVE) (2020-Present)	3.7-4.2 m	8 Bands			0.431-0.885	

SkySat Camera

Sensor Name	Vehicle		Resolution				Wavelength (μm)	Swath width (km)
			Spatial	Spectral	Temporal	Radiometric		
SkySat Camera (USA)	SkySat Contellation (2013-Present) (21 Satellites) No. of Sensors: 03	SkySat 1-2	Pan; 0.86m MS; 1m	5 Bands	4-5 Days	16 Bits	0.45-0.9	8
		SkySat 3-15	Pan; 0.65m MS; 0.81m					5.9
		SkySat 16-21	Pan; 0.57m MS; 0.75m					5.5

GeoEye Imaging System

Sensor Name	Vehicle	Resolution				Wavelength (μm)	Swath width (km)
		Spatial	Spectral	Temporal	Radiometric		
GeoEye Imaging System (USA)	GeoEye-1 (2008-Present)	Pan; 0.41m MS; 1.64m	5 Bands	≤ 3 days	11 Bits	0.45-0.92	15.2

High Resolution Imager (HiRI)

Sensor Name	Vehicle	Resolution				Wavelength (μm)	Swath width (km)
		Spatial	Spectral	Temporal	Radiometric		
High Resolution Imager (HiRI) (France)	Pleiades 1A (2011-Present)	Pan; 0.5m MS; 2m	5 Bands	1 Day	12 Bits	0.48-0.94	20-53
	Pleiades 1B (2012-Present)						

Application of Sensors

Name of Sensors	Special Applications	Common Applications
PlanetScope Camera	• Mining: land reclamation, • Slope monitoring	1. Agriculture – Crop health monitoring, precision farming, yield estimation. 2. Forestry – Deforestation detection, forest health and biomass assessment. 3. Environmental Monitoring – Tracking floods, droughts, pollution, and land degradation. 4. Disaster Monitoring/Management – Assessing and mapping natural disasters like floods, landslides, and fires. 5. Land Use / Land Cover Mapping – Identifying changes in urban areas, vegetation, and surface features.
SkySat Camera	• Urban planning, • Infrastructure monitoring, • Defense & security,	
GeoEye Imaging System	• High-precision mapping & surveying, • Defense & intelligence operations.	
High-Resolution Imager	• Geological mapping, • Disaster management	
Remote Sensing Instrument	• Urban/regional planning, • Water resources management, • Detailed mapping	
Vegetation Monitoring Instrument	• Crop yield prediction, • Drought detection, • Vegetation health monitoring	



Thank you for your attention